

NEWQUAY CORNWALL, United Kingdom

WMO: 033465

Lat: 50.433N

Lon: 4.983W

Elev: 119

StdP: 99.90

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.7	0.6	-4.8	2.6	1.4	-3.4	2.9	2.3	16.7	9.2	15.2	8.0	4.4	70	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	23.0	17.3	21.2	16.5	19.8	16.0	18.4	21.3	17.6	19.8	16.9	18.7	5.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.7	19.0	16.7	12.1	18.3	16.1	11.6	17.7	52.5	21.4	49.9	19.9	47.7	18.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.9	11.5	-2.5	25.6	1.6	2.1	-3.7	27.1	-4.6	28.3	-5.5	29.5	-6.7	31.0
			-3.6	19.7	1.7	1.5	-4.8	20.7	-5.8	21.6	-6.8	22.4	-8.0	23.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	6.9	6.9	7.7	9.1	11.8	14.2	16.0	16.5	14.7	12.3	9.5	7.3	(6)
(7)		DBStd	4.14	2.83	2.51	2.27	2.30	2.30	1.98	1.93	1.79	1.72	2.06	2.15	2.88	(7)
(8)		HDD10.0	439	98	89	74	42	8	0	0	0	0	6	34	88	(8)
(9)		HDD18.3	2652	353	321	328	278	202	127	77	63	108	187	266	341	(9)
(10)		CDD10.0	843	3	2	4	15	64	125	186	201	142	77	19	6	(10)
(11)		CDD18.3	14	0	0	0	0	0	2	6	6	1	0	0	0	(11)
(12)		CDH23.3	49	0	0	0	0	0	9	18	21	1	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	1	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	5.6	6.9	6.5	6.1	5.4	5.1	4.9	4.6	4.6	5.0	6.0	5.9	6.6	(14)
(15)	Precipitation	PrecAvg	1124	123	104	77	76	69	67	70	74	79	123	132	132	(15)
(16)		PrecMax	1397	220	211	140	168	142	130	147	178	170	226	273	282	(16)
(17)		PrecMin	895	27	24	21	16	7	17	7	14	24	50	55	35	(17)
(18)		PrecStd	167	52	54	35	37	40	38	37	46	40	43	57	63	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	12.7	15.0	18.2	21.8	24.6	25.6	25.9	22.6	18.0	15.1	13.2	(19)
(20)			MCWB	11.3	11.2	10.4	12.1	15.1	17.6	18.5	18.8	17.5	15.0	13.9	12.3	(20)
(21)		2%	DB	12.1	11.7	12.7	15.5	19.4	21.0	22.9	22.6	20.1	16.6	13.9	12.6	(21)
(22)			MCWB	11.1	10.6	10.1	11.3	14.0	15.6	17.2	17.9	16.2	14.5	12.8	11.8	(22)
(23)		5%	DB	11.3	10.9	11.7	13.7	17.3	19.1	21.0	20.6	18.6	15.8	13.3	12.0	(23)
(24)			MCWB	10.4	9.7	10.0	10.5	13.3	14.9	16.6	16.9	15.5	14.2	12.2	11.1	(24)
(25)		10%	DB	10.5	10.2	10.8	12.4	15.4	17.5	19.2	19.3	17.6	15.2	12.6	11.3	(25)
(26)			MCWB	9.6	9.1	9.5	10.0	12.4	14.3	15.9	16.5	15.0	13.9	11.5	10.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	12.0	11.7	13.4	16.1	18.4	19.1	19.8	18.4	16.1	14.4	12.7	(27)
(28)			MCDWB	12.2	12.5	12.6	15.8	20.0	23.6	24.0	22.9	21.0	16.8	14.8	12.9	(28)
(29)		2%	WB	11.3	11.0	11.0	12.0	14.7	16.6	17.9	18.8	17.2	15.3	13.4	12.1	(29)
(30)			MCDWB	11.8	11.5	12.0	14.3	17.9	19.5	20.9	21.1	18.9	16.0	13.7	12.5	(30)
(31)		5%	WB	10.6	10.2	10.4	11.2	13.8	15.6	17.1	17.9	16.5	14.7	12.6	11.4	(31)
(32)			MCDWB	11.2	10.6	11.3	12.9	16.3	17.8	19.5	19.7	17.8	15.4	13.1	11.9	(32)
(33)		10%	WB	9.9	9.4	9.8	10.4	12.9	14.8	16.5	17.1	15.8	14.2	11.7	10.6	(33)
(34)			MCDWB	10.5	10.0	10.6	11.9	14.7	16.7	18.5	18.7	17.0	15.0	12.4	11.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.3	4.6	5.3	5.6	5.2	5.0	5.2	4.5	4.0	4.1	(35)	
(36)			MCDBR	4.1	4.3	5.6	7.2	8.5	7.9	8.2	7.1	6.7	4.8	4.0	3.9	(36)
(37)			MCWBR	4.2	3.9	3.9	4.4	5.0	4.5	4.3	3.8	4.0	3.7	3.9	4.5	(37)
(38)		5% WB	MCDWB	4.0	3.9	4.5	5.9	7.2	6.7	6.5	6.1	5.6	4.3	3.5	3.9	(38)
(39)			MCWBR	4.3	4.1	3.9	4.1	4.5	4.2	3.9	3.7	4.1	3.8	4.0	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.345	0.371	0.379	0.387	0.374	0.378	0.363	0.360	0.353	0.331	0.324	(40)	
(41)		taud	2.364	2.392	2.335	2.340	2.358	2.423	2.417	2.477	2.474	2.456	2.412	2.329	(41)	
(42)		Ebn at Noon	688	785	827	863	870	884	873	871	835	767	688	632	(42)	
(43)		Edh at Noon	66	82	103	114	117	110	109	98	89	76	62	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.80	1.60	2.74	4.41	5.35	5.89	5.38	4.60	3.46	1.96	1.03	0.64	(44)	
(45)		RadStd	0.07	0.15	0.30	0.29	0.36	0.53	0.38	0.30	0.23	0.13	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page